I. GENESIS · SENSORY DATA

01

A Sound Instead of a Buzzword

Most resumes begin with a dense list of corporate buzzwords. Let me begin with a sound:

The deep mechanical hum of MareNostrum 5 calculating climate turbulence across thousands of GPUs, or the gentle chime of a motion-sensing sphere when my cat Pipa knocks it across the living room floor.

Data is not just an analytical output to be confined to rows and columns. Inspired by Domestic Data Streamers, I believe data is a living physical material—something to be felt, touched, heard, and inhabited.

Modern science is uncovering the fundamental mechanics of our world, but when those breakthroughs remain trapped behind cold cluster logs and impenetrable matrices, we have only completed half the mission.

My conviction is the democratization of science: turning invisible computational realities into unforgettable, sensorial human stories that touch non-specialists.

II. RIGOUR · EXTREME COMPUTE

02

From First-Principles to Tier-0

My creative impulses are anchored in genuine first-principles physics and high-performance software rigour:

- Double Degree in Physics & Mathematics
  (Univ. of Seville, 2013–2018 · 8.35/10 · 3rd of Class)
- MSc in Plasma & Fusion Physics
  (Aix-Marseille Univ., 2019–2020 · 15.43/20 · 1st of Class)

- BSC Performance Team (May 2025 – Present):
  Contributing to ocean-atmosphere coupling in European digital twins (DestinationE & ClimateDT); ECLand model advancement; deploying workflows on MareNostrum 5.

- Fusion Team & CENER Background:
  Optimizing Tier-0 fusion solvers (OpenMP, MPI, PETSc, OpenACC, oneAPI); ALYA-MAGNET modeling; building real-time predictive wind plant telemetry pipelines.

- AI-Assisted Engineering:
  I treat LLMs as fluid thought-partners, bouncing rapidly from C++/Fortran HPC kernels to browser sound design, motion graphics, and hardware firmware.

III. STUDIO · PHYSICAL COMPUTING

03

Four Creative Experiments

MNusic — Listening to & Visualizing MN5 & LUMI

- Sonifies cluster power, GPU load & thermal gradients into harmonic ambient soundscapes (Web Audio API) & particles.
- Transforms abstract petascale algorithms into audible life.
- Stack: HPC Telemetry, Web Audio, HTML5 Canvas, AI pairing.

Pipa-Music-Ball — Bio-Kinetic Interactive Artifact

- Embedded physical sphere (ESP32, 6-DoF IMU) that turns cat kinetic play into microtonal scales & acoustic chimes.
- Tangible proof of non-anthropocentric playful data design.
- Stack: Microcontrollers, I2S DAC, inertial sensing, 3D body.

Expo-Curator — github.com/carrommad/expo-curator

- Image curation, automated sub-classification, booklet design, and spatial exhibition wall layout generator for galleries.
- Directly serves the staging needs of data art exhibitions.
- Stack: Python/JS, layout geometry, automated taxonomy.

Lens & Motion — Photography & Adobe After Effects

- Still photography composition (framing, geometric light) paired with procedural motion graphics & kinetic typography.
- Turns scientific findings into broadcast-grade video reels.
- Stack: Adobe After Effects, color grading, visual rhythm.

IV. CONVERGENCE · FUTURE ACTION

04

Synergy with DAV & Fernando Cucchietti

Following your workshops and tracking your flagship initiatives reveals an organic convergence with my skills and passion:

- GDU-HPC (Gemelo Digital Urbano):
  Direct synergy with my HPC digital twin workflow at BSC (DestinationE & ClimateDT) and large-scale urban analytics.

- AIR (Air for European Resilience):
  Connecting fluid/atmospheric physics (MSc Plasma & Fluid) with participatory public environmental storytelling.

- REALLOCATE (Climate-Friendly Active Mobility):
  Data visualization for rethinking street design & emissions.

Your 5 core lines (Scientific Viz, Interactive Viz, Industrial AI, Creative Intelligence, Urban Data) are exactly where I belong.

I don't just want to watch from the audience—I want to join Fernando Cucchietti and the team to engineer the most daring, memorable scientific data installations in the world.

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